

Enhancing Debris Flow Warning with Deep Machine Learning

Abstract

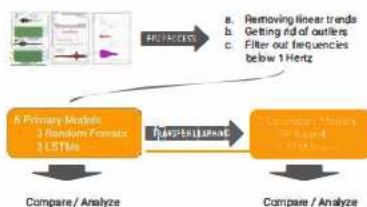
This study aims to address the following research question: How do deep learning methods compare to traditional methods at optimizing debris flow detection across multiple geographical areas? To do this, this paper will gather time series seismic datasets from three geographical distinct areas, train deep and traditional machine learning models on these datasets, and compare how well deep learning models compare to traditional methods. These datasets will go through a standardized preprocessing pipeline for time series data. The Waveforms will be resampled to a common frequency, with a high pass filter (e.g. filtering out all noise below 1 Hertz). Model training will proceed in parallel for the deep learning and traditional methods. For the deep learning methods, three RNNs will be tested. For the traditional methods, three random forests will be tested. After training individual models on each dataset, transfer learning and model fusion will be employed to create two consolidated models. The deep learning consolidated model will be built by combining the RNNs through stacking learned embeddings into a small fusion network. Then transfer learning will be used to fine tune the model on a created test dataset. This test dataset is a compilation of datapoints from each of the other datasets. The traditional learning consolidated model will also be built by combining the Random Forests through stacking learned embeddings, and then fine tuning it with transfer learning and the test dataset.

Introduction

Debris flows are mixtures of water, sediment, and rock that rapidly fall down a mountainside — often due to gravity. In many ways debris flow events act like a flowing fluid, due to the high water content. However, the solid components they carries ranging from fine particles to massive boulders make them extremely destructive (USGS, 2025). Debris flow typically occur on steep slopes, such as hillsides and mountains; they are triggered by intense rainfall, rapid snowmelt, wildfires, or any other change that vastly increases the water levels or reduce the slope stability (Landslide Hazards Program, 2018). Typical speeds of debris flow events often exceed 35 mph. Due to these high speeds and high mass content, debris flow events may even bury terrain with thick layers of sediment and rock. The main danger of debris flow events, however, does not stem from their destructive capabilities, but instead stems from their unpredictability. It is often not possible to precisely anticipate when and exactly where debris flow will occur. Nor is it possible to determine how large they will become or what specific path they will take. Moreover, due to the steep terrain they reside over, the flow of rocks and boulders could quickly gain momentum, jump out of artificial stream channels, overflow banks, and impact areas miles away from the origin. It's large area of effect can cause havoc on people and infrastructure downstream especially if these events are triggered by short intense storms (NWS, 2023).

Methodology

This study aims to address the following research question: How do deep learning methods compare to traditional methods at optimizing debris flow detection across multiple geographical areas? Previous work has shown that machine learning can substantially improve debris flow warning at specific geographic locations. However most literature focuses on one location, and they typically rely on traditional machine learning algorithms, such as Random Forests or Support Vector Machines (Chmiel et al., 2021). While these have shown some promise, they have struggled with generalizability; that is they struggle to generalize across different environments (Shou et al., 2025). Therefore, it remains to be seen whether deep learning architectures, particularly recurrent Neural Networks (RNNs), can outperform traditional methods in a more diverse geographical dataset. To do this, this paper will gather time series seismic datasets from three geographical distinct areas, train deep and traditional machine learning models on these datasets, and compare how well deep learning models compare to traditional methods. For clarity, terms and variable will be defined as follows. A debris flow event refers to a continuous time interval in the seismic time-series data where ground motion is completely or almost completely caused by debris flow events. These intervals have already been designated by other literature. Lead time is the interval between when a model predicts a debris flow event and when a significant amount of debris hits a specified location downstream. A true positive occurs when the machine learning model predicts an event during the predetermined debris flow interval. The independent variable is the type of machine learning model (i.e. a RNN or Random Forest) The dependent variables include the metrics of the machine learning models, including precision, sensitivity, F1 score, ROC-AUC, and median lead time for true positives. The paper focuses on time-series seismic data from three sources. The first dataset comes from the Swiss Seismological Service. This dataset has been used extensively in past literatures monitoring the Illgraben catchment (Wenner et al., 2021). The second dataset comes from the Hydroshare Repository, which provides open access hydrological and geographical data from areas at risk of debris-flow events. The third dataset comes from the Southern California Seismic Network, which catalogues the 2018 Montecito debris-flow event. These three datasets — spanning the Alps, US mountains, and Coastal California — directly address the limitations noted in prior studies: the fact that models were typically trained and tested on a dataset from a single geographic location (Chmiel et al., 2021). These datasets will go through a standardized preprocessing pipeline for time series data. The Waveforms will be resampled to a common frequency, with a high pass filter (e.g. filtering out all noise below 1 Hertz). The seismic data will then be segmented into overlapping windows of 100 second. From these segments, raw data and engineered features — such as waveform statistics, spectral features, spectrogram metrics, and network correlations — will be derived. Model training will proceed in parallel for the deep learning and traditional methods. For the deep learning methods, three RNNs will be tested. For the traditional methods, three random forests will be tested. After training individual models on each dataset, transfer learning and model fusion will be employed to create two consolidated models. The deep learning consolidated model will be built by combining the RNNs through stacking learned embeddings into a small fusion network. Then transfer learning will be used to fine-tune the model on a created test dataset. This test dataset is a compilation of datapoints from each of the other datasets. The traditional learning consolidated model will also be built by combining the Random Forests through stacking learned embeddings, and then fine tuning it with transfer learning and the test dataset. The novelty of this method is that it combines multiple geographical site data and the integration of deep learning in this machine learning pipeline. Once again, while past research has demonstrated that machine learning can detect debris flows, these studies have not approached the subject of generalizing these machine learning algorithms to other geographical occasions. By explicitly testing RNN-based models trained across a larger area, this study address the most demanding gaps in debris flow monitoring research.



Results

	BASE MODELS		TRANSFER LEARNING	
	Random Forest	LSTM	Random Forest	LSTM
Illgraben	91%	98%	93%	97%
Gorkha	81%	95%		
South Cali	85%	95%		

The paper focuses on time series seismic data from three sources. The first dataset comes from the Swiss Seismological Service. This dataset has been used extensively in past literatures monitoring the Illgraben catchment (Wenner et al., 2021). The second dataset comes from the Hydroshare Repository, which provides open-access hydrological and geographical data from areas at risk of debris-flow events. The third dataset comes from the Southern California Seismic Network, which catalogues the 2018 Montecito debris flow event. These three datasets — spanning the Alps, US mountains, and Coastal California — directly address the limitations noted in prior studies: the fact that models were typically trained and tested on a dataset from a single geographic location (Chmiel et al., 2021). These datasets will go through a standardized preprocessing pipeline for time series data. The Waveforms will be resampled to a common frequency, with a high pass filter (e.g. filtering out all noise below 1 Hertz). The seismic data will then be segmented into overlapping windows of 100 second. From these segments, raw data and engineered features — such as waveform statistics, spectral features, spectrogram metrics, and network correlations — will be derived. Model training will proceed in parallel for the deep learning and traditional methods. For the deep learning methods, three RNNs will be tested. For the traditional methods, three random forests will be tested. After training individual models on each dataset, transfer learning and model fusion will be employed to create two consolidated models. The deep learning consolidated model will be built by combining the RNNs through stacking learned embeddings into a small fusion network. Then transfer learning will be used to fine-tune the model on a created test dataset. This test dataset is a compilation of datapoints from each of the other datasets. The traditional learning consolidated model will also be built by combining the Random Forests through stacking learned embeddings, and then fine tuning it with transfer learning and the test dataset. The novelty of this method is that it combines multiple geographical site data and the integration of deep learning in this machine learning pipeline. Once again, while past research has demonstrated that machine learning can detect debris flows, these studies have not approached the subject of generalizing these machine learning algorithms to other geographical occasions. By explicitly testing RNN-based models trained across a larger area, this study address the most demanding gaps in debris flow monitoring research.

Conclusion

The conclusion for my study is that Deep Learning models do have a significant impact on the accuracy of Debris flow warning classifications. The base random forest models showed accuracies ranging from 81% to 91%. However the corresponding debris flow models that use deep learning have accuracies ranging in the high 90s (95% to 98%). This increase in accuracies shows that deep learning is better for debris classification than traditional machine learning models.

Additionally, the transfer learning models generally had high accuracies than the base models for that the transfer learning models derived from. For the random forest models, each accuracy ranged from 13% to 2% lower than the accuracy of the transfer learning model. Similarly, for the long-short term memory model, it generally had a higher accuracy than the base models. However, for the Illgraben data set, it had a lower accuracy. This could suggest that the generalization method (in this case transfer learning) may not carry as much weight, so other methods of generalizing the model to multiple geographic locations should be tested.

Acknowledgements

- Chen, T.-L., Wu, Y.-H., & Chiu, Y. H. (2025). Debris flow risk characteristics and potential spatial mitigation strategies under extreme rainfall events. *Climate Services*, 40, 100618. <https://doi.org/10.1016/j.clis.2025.100618>
- Chmiel, M., Walter, F., Wenner, M., Zhai, Z., Alsdorf, B. W., & Hober, C. (2021). Machine learning improves debris flow warning. *Geophysical Research Letters*, 48(18), e2020GL090874. <https://doi.org/10.1029/2020GL090874>
- Dahlgren, M. P., & West, A. J. (2019). In situ and run-out of post-seismic debris flows: Insights from the 2015 Gorkha earthquake. *Geophysical Research Letters*, 46(18), 9859–9868. <https://doi.org/10.1029/2019GL083549>
- Ivanov, I. (2024). *Hyvry Examples Example: ether: traditional pytorch main - pytorch se/HASRPE GitHub*. <https://github.com/pytorch/examples/blob/master/imagenet/main.py>
- Koering, A., Hübl, J., Sui, H., E., V. H., & McNeil, B. W. (2014). Infrasound produced by debris flow: Propagation and frequency content evolution. *Natural Hazards*, 70(3), 1713–1733. <https://doi.org/10.1007/s11069-011-0741-8>